# Traffic Sign Classification Under Varying Environmental Conditions

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***Abstract*—Traffic sign classification is important for intelligent transportation systems, but environmental conditions can reduce recognition performance. This study evaluates traffic sign classification using the German Traffic Sign Recognition Benchmark (GTSRB) dataset and features extracted by a pretrained DINOv2 model. Two classifiers, a Multi-Layer Perceptron (MLP) and a Linear Support Vector Machine (SVM), are compared using an 80:20 training-validation split. Robustness is examined under 14 types of image corruption, each applied at five severity levels. Performance is measured using accuracy and macro F1-score. On the validation dataset, MLP achieves 96.30% accuracy and a macro F1-score of 0.9656, while Linear SVM achieves 90.67% accuracy and a macro F1-score of 0.9203. The corruption experiments further examine how different image distortions affect classification performance.**

***Index Terms*—Traffic sign classification, image corruption, robustness, DINOv2, multilayer perceptron, support vector machine**

## I. Introduction

Traffic sign recognition plays an important role in self-driving cars and intelligent transportation systems. It helps vehicles identify road signs and follow traffic rules. However, recognizing traffic signs becomes more difficult when images are affected by real-world conditions such as fog, snow, poor lighting, blur, and noise.

This project studies how these conditions affect traffic sign classification using the GTSRB dataset. A pretrained DINOv2 model is used to extract features from traffic sign images, which are then classified using MLP and Linear SVM models. The models are evaluated on clean images and 14 types of image corruption at five severity levels. Accuracy and macro F1-score are used to compare their performance and understand how well they handle different levels of corruption.

## II. Related Work

Traffic sign recognition has been studied using traditional machine learning and deep learning techniques. The GTSRB dataset is widely used to evaluate traffic sign classification models. More recent approaches, such as pretrained vision models, can extract useful image features and reduce the need to train a model from scratch.

DINOv2 is a pretrained vision model that learns visual features from images. These features can be used with different classifiers for image recognition tasks. However, image corruptions such as noise, blur, fog, and changes in brightness can affect classification performance. This study focuses on evaluating DINOv2-based features with MLP and Linear SVM classifiers under different corruption types and severity levels.

## III. Proposed System Architecture

The proposed system evaluates traffic sign classification under different image conditions using features extracted from a pretrained vision model. The GTSRB dataset is first loaded and divided into training and validation sets. The images are preprocessed and passed through DINOv2-Small to obtain 384-dimensional feature vectors.

These features are used to train two classifiers, namely MLP and Linear SVM. Their performance is initially measured on clean validation images. To examine robustness, 14 types of image corruption are applied at five severity levels. The corrupted images are then evaluated using the trained classifier, and the results are recorded using accuracy and macro F1-score.

Finally, the results are analyzed using performance graphs and confusion matrices. This workflow allows the study to compare clean-image performance with performance under different types and levels of image degradation.

## IV. Dataset and Methodology

### A. Dataset

The German Traffic Sign Recognition Benchmark (GTSRB) dataset is used to study traffic sign classification under different image conditions. It contains images belonging to 43 traffic sign categories. Before feature extraction, the images are resized to 224 × 224 pixels and normalized. The dataset is divided into training and validation sets using a stratified 80:20 split with random seed 42. This division maintains a similar distribution of traffic sign classes in both sets. The validation set contains 7,842 images and is used to measure the performance of the trained classifiers.

### B. Feature Extraction

This study uses a pretrained DINOv2-Small model to obtain visual features from traffic sign images. Instead of training an image recognition network from scratch, the model is used to convert each image into a 384-dimensional feature vector.

These vectors capture visual patterns that help distinguish one traffic sign category from another. The extracted features are saved and reused during classifier training and testing, reducing the need to process the same images repeatedly through the vision model.

### C. Classification Models

Two classification methods are considered: a Multi-Layer Perceptron (MLP) and a Linear Support Vector Machine (SVM). Both models receive the feature vectors generated by DINOv2. The MLP uses fully connected layers with 256 and 128 hidden units, ReLU activation functions, and dropout to reduce dependence on individual neurons during training. Its output layer predicts one of the 43 traffic sign classes. The Linear SVM provides a second classification approach for comparison. Evaluating both models on the same extracted features helps determine how the choice of classifier affects recognition performance.

### D. Image Corruptions

Traffic sign images can lose important visual details because of changes in weather, lighting, image quality, and camera movement. To examine these effects, this study considers 14 corruption types, each with five severity levels. The selected corruptions are Gaussian noise, impulse noise, shot noise, snow, fog, frost, motion blur, zoom blur, defocus blur, brightness changes, contrast changes, pixelation, elastic transformation, and JPEG compression. Each corruption is applied to the validation images at severity levels from 1 to 5. The resulting predictions are compared with the clean-image baseline to measure how image degradation affects classification performance.

## V. Experimental Setup

The GTSRB dataset is divided into training and validation sets using an 80:20 ratio. A pretrained DINOv2-Small model extracts 384-dimensional feature vectors from the traffic sign images. These features are used to train two classifiers: a Multi-Layer Perceptron (MLP) and a Linear Support Vector Machine (SVM).

The classifiers are evaluated on clean validation images to establish baseline performance. Robustness is then examined using 14 types of image corruption, each applied at five severity levels, resulting in 70 corrupted conditions. Accuracy and macro F1-score are used to measure classification performance and compare the effect of different corruptions.

The project is implemented in Python, with PyTorch used for feature extraction and MLP training. The extracted features and trained models are saved for reuse during evaluation. Separate scripts handle dataset preparation, feature extraction, model training, corruption generation, and performance analysis. The evaluation results are stored in a CSV file, while graphs and confusion matrices are generated to help interpret the experimental findings.

## VI. Evaluation Metrics

The performance of the classifiers is evaluated using accuracy and macro F1-score. Accuracy measures the percentage of correctly classified images, while macro F1-score gives equal importance to each traffic sign class.

Accuracy is calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100 \tag{1}$$

Precision and recall are used to calculate the F1-score:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The F1-score is calculated as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

The macro F1-score is obtained by averaging the F1-scores of all classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{5}$$

Here, $TP$ represents true positives, $FP$ represents false positives, $FN$ represents false negatives, and $C$ is the total number of classes.

## VII. Results and Discussion

The performance of the MLP and Linear SVM classifiers is evaluated on clean and corrupted traffic sign images. The experiments examine the classification performance of both models and investigate how different image corruptions affect recognition accuracy.

### A. Clean-Image Classification Performance

Both classifiers are first evaluated on clean validation images to establish a baseline for the corruption experiments. The MLP achieves a validation accuracy of 96.30% and a macro F1-score of 0.9656, while the Linear SVM achieves 90.67% accuracy and a macro F1-score of 0.9203. These results indicate that the MLP performs better than the Linear SVM on clean images.

The effect of image corruptions is then analyzed using accuracy and macro F1-score results across different corruption types and severity levels. Confusion matrices are also examined to understand the classification errors made by each model.

### B. Training Performance and Generalization

The training and validation results provide an overview of how well the classifiers learn from the extracted features and perform on unseen images. The MLP achieves a training accuracy of 99.38% and a validation accuracy of 96.30%. Its training and validation macro F1-scores are 0.9956 and 0.9656, respectively. The Linear SVM obtains a training accuracy of 96.43% and a validation accuracy of 90.67%, with corresponding macro F1-scores of 0.9816 and 0.9203.

Both classifiers perform better on the training data than on the validation data. The difference is larger for the MLP in absolute accuracy terms, although this alone does not establish severe overfitting. The results indicate that validation performance should be considered alongside training performance when comparing the two classifiers.

### C. Comparison of Classification Models

The MLP model achieves 96.30% validation accuracy with a macro F1-score of 0.9656. The Linear SVM model achieves 90.67% validation accuracy with a macro F1-score of 0.9203. These results show that MLP performs better on the validation dataset.

### D. Effect of Image Corruptions

The evaluation shows that different types of image corruption affect classification performance differently. Defocus blur and pixelation have relatively high average accuracy, while frost, snow, fog, and noise cause greater performance drops. This indicates that the type of corruption plays an important role in traffic sign recognition.

### E. Comparison Across Corruption Types

The average results across five severity levels show that different corruptions have different effects on traffic sign classification. Defocus blur achieves the highest average accuracy at 91.07%, followed closely by pixelation at 91.00%. Motion blur and brightness changes also produce comparatively better results than several other corruption types.

In contrast, frost produces the lowest average accuracy at 26.69%. Snow and fog achieve average accuracies of 36.29% and 38.37%, respectively. Noise-based corruptions also reduce performance, with Gaussian noise achieving 44.55% average accuracy, shot noise 48.20%, and impulse noise 38.70%.

These findings indicate that the classifier is more sensitive to some image distortions than others. Corruptions that substantially change the appearance or visibility of traffic signs can make classification more difficult. The macro F1-score results provide additional information about performance across classes and should be considered alongside accuracy.

### F. Performance Across Severity Levels

The effect of corruption severity is examined by evaluating the MLP classifier at five levels for each corruption type. The results show that stronger image degradation can reduce classification accuracy. For example, under Gaussian noise, accuracy decreases from 68.77% at severity level 1 to 17.21% at severity level 5. This indicates that severe noise makes it more difficult for the classifier to identify traffic signs correctly.

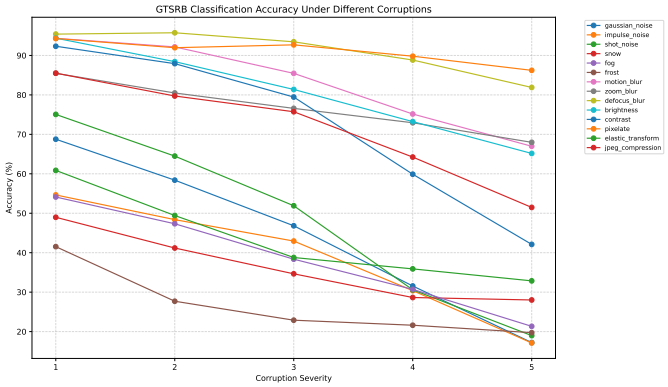



Fig. 1. MLP classification accuracy across corruption severity levels.

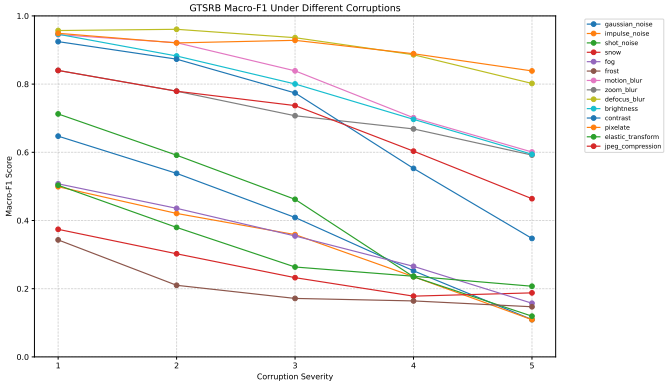



Fig. 2. MLP macro F1-score across corruption severity levels.

However, the reduction in performance is not necessarily uniform across all corruption types and severity levels. Some variations may occur depending on how each corruption affects the visual features of an image. Fig. 1 presents the changes in classification accuracy across severity levels, helping identify the conditions under which the classifier becomes less reliable.

### G. Confusion Matrix Analysis

The confusion matrices provide a class-wise view of the classification results on clean validation images. They show which traffic sign classes are correctly classified and which classes are more frequently confused with one another. Figures 3 and 4 show the confusion matrices for the Linear SVM and MLP classifiers, respectively.

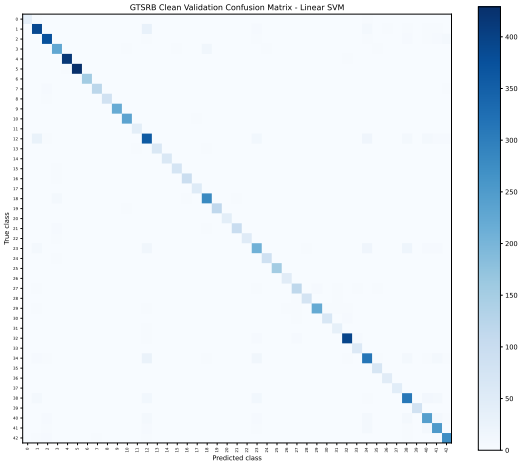



Fig. 3. Confusion matrix for clean-image classification using Linear SVM.

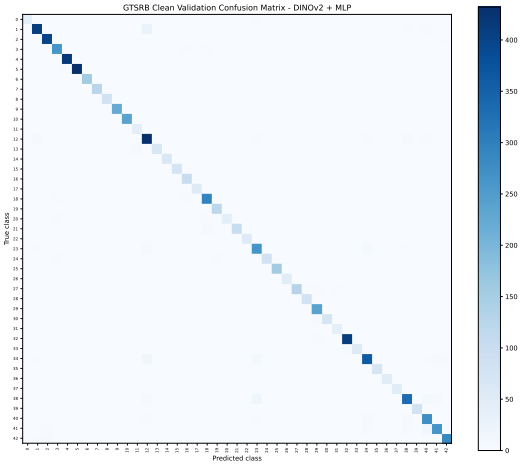



Fig. 4. Confusion matrix for clean-image classification using MLP.

## VIII. Conclusion

This study evaluates traffic sign classification using pre-trained DINOv2 features with MLP and Linear SVM classifiers. The MLP achieves higher validation accuracy than the Linear SVM on the GTSRB dataset. The experiments also show that different image corruptions affect classification performance differently. Conditions such as frost, snow, fog, and noise create greater challenges for recognition. Overall, this study highlights the importance of evaluating traffic sign classification models under different environmental conditions.

## References

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